Inequality Proofs

Math 742

1/16/2026

1. Markov's Inequality

Theorem (Markov). Let $X \geq 0$ a.s. with $\mathbb{E}[X] < \infty$. Then for any $t > 0$,

$$\mathbb{P}(X \geq t) \leq \frac{\mathbb{E}[X]}{t}.$$

Proof.

$$\mathbb{E}[X] = \int_0^\infty x\,dF(x) \geq \int_t^\infty x\,dF(x) \geq \int_t^\infty t\,dF(x) = t\,\mathbb{P}(X \geq t).$$

Dividing by $t > 0$ yields the result. $\square$

2. Chebyshev's Inequality

Theorem (Chebyshev). If $\mathbb{E}[X] = \mu$ and $\mathbb{V}(X) = \sigma^2 < \infty$, then

$$\mathbb{P}(|X - \mu| \geq t) \leq \frac{\sigma^2}{t^2}.$$

Proof. Apply Markov to the non-negative r.v. $(X - \mu)^2$:

$$\mathbb{P}(|X - \mu| \geq t) = \mathbb{P}((X - \mu)^2 \geq t^2) \leq \frac{\mathbb{E}[(X - \mu)^2]}{t^2} = \frac{\sigma^2}{t^2}.$$

$\square$

3. Cauchy–Schwarz Inequality (for expectations)

Theorem. For random variables X, Y with finite second moments,

$$|\mathbb{E}[XY]|^2 \leq \mathbb{E}[X^2]\mathbb{E}[Y^2].$$

Proof. For any $\lambda \in \mathbb{R}$,

$$0 \leq \mathbb{E}[(\lambda X + Y)^2] = \lambda^2\mathbb{E}[X^2] + 2\lambda\mathbb{E}[XY] + \mathbb{E}[Y^2].$$

The quadratic in λ has non-positive discriminant:

$$(2\mathbb{E}[XY])^2 - 4\mathbb{E}[X^2]\mathbb{E}[Y^2] \leq 0 \;\Rightarrow\; |\mathbb{E}[XY]|^2 \leq \mathbb{E}[X^2]\mathbb{E}[Y^2].$$

$\square$

Corollary. $|\mathrm{Cov}(X, Y)|^2 \leq \mathrm{Var}(X)\mathrm{Var}(Y)$.

4. Jensen's Inequality

Theorem (Jensen). Let f be convex on an interval I. Then for any r.v. X with range in I,

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)].$$

If f is strictly convex, equality holds iff X is degenerate.

Proof (for twice-differentiable case). By Taylor expansion with remainder,

$$f(x) = f(\mu) + f'(\mu)(x-\mu) + \frac{f''(c_x)}{2}(x-\mu)^2$$

for some c_x between x and $\mu = \mathbb{E}[X]$. Taking expectation:

$$\mathbb{E}[f(X)] = f(\mu) + 0 + \mathbb{E}\left[\frac{f''(c_X)}{2}(X-\mu)^2\right] \geq f(\mu)$$

since $f'' \geq 0$ for convex f. $\square$

Key Application in Math 742.

$f(t) = -\log t$ is strictly convex on $(0,\infty)$ ($f''(t) = 1/t^2 > 0$).
For $0 < p < 1$,

$$p\log p + (1-p)\log(1-p) \leq \log(p \cdot p + (1-p)\cdot(1-p)) = \log 1 = 0,$$

with equality **never** in the interior.

Thus the binomial log-likelihood is always negative for $0 < S < n$.

5. Hoeffding's Inequality

Theorem (Hoeffding 1963). Let $X_1, \ldots, X_n$ be independent with $a_i \leq X_i \leq b_i$ a.s. Then

$$\mathbb{P}\left(\left|\frac{1}{n}\sum X_i - \mathbb{E}\left[\frac{1}{n}\sum X_i\right]\right| \geq \varepsilon\right) \leq 2\exp\left(-\frac{2n^2\varepsilon^2}{\sum_{i=1}^n (b_i-a_i)^2}\right).$$

Special Case – Bernoulli(p): $X_i \in \{0,1\}$, so $b_i - a_i = 1$ for all i. Hence

$$\boxed{\mathbb{P}(|\bar{X}_n - p| \geq \varepsilon) \leq 2\exp(-2n\varepsilon^2)}.$$

Proof Idea (Hoeffding's Lemma + Chernoff).

For a bounded r.v. $Z \in [a,b]$, $\mathbb{E}[Z] = \mu$,

$$\mathbb{E}[e^{t(Z-\mu)}] \leq \exp\left(\frac{t^2(b-a)^2}{8}\right).$$

Apply to $S_n = \sum(X_i - \mu_i)$, use independence, and optimize the Chernoff bound over $t > 0$. Full proof in Wasserman Ch 3.

6. Mills' Inequality (Gaussian tail bound)

Theorem. For $Z \sim N(0,1)$ and $t > 0$,

$$\frac{e^{-t^2/2}}{t\sqrt{2\pi}}\left(1 - \frac{1}{t^2}\right) < \mathbb{P}(Z > t) < \frac{e^{-t^2/2}}{t\sqrt{2\pi}}.$$

Proof (integration by parts). Let $\phi(t) = e^{-t^2/2}/\sqrt{2\pi}$. Then

$$\mathbb{P}(Z > t) = \int_t^\infty \phi(u)\,du.$$

Integrate by parts with $u = 1/u$, $dv = u\phi(u)\,du$ to get the upper bound. The lower bound follows similarly with an extra term. $\square$

Summary of What You Must Know for Exams

- Markov $\to$ Chebyshev $\to$ (Weak Law of Large Number (WLLN))
- Cauchy–Schwarz $\to$ Correlation bounded by 1
- Jensen + $-\log t$ convex $\to$ Binomial log-likelihood ≤ 0 (strict)
- Hoeffding $\to$ Exponential concentration for bounded r.v.s (especially Bernoulli)
- Mills $\to$ Sub-Gaussian tails for normal

These six inequalities are the **only ones** used in proofs throughout Math 742.